

Giri's Tech Hub Pvt.Ltd, Pune.
Programming (Machine) Test

Batch: Sep-24/Oct-24

Date: 30/01/2025
Time: 08:00 to 10.00 Am

Instructions:

Total:- 10 Marks

1. Solve any six questions.
2. Input should be from user.
3. Indentation and comments mandatory.
4. Each program 1.5 Marks and 1 Marks Comments.

Q1. Write a java program to print this pattern.

```
A B C D E
  A B C D
    A B C
      A B
        A
      A B
    A B C
  A B C D
A B C D E
```

Q2. Write a java program to calculate the sum of 1 to nth prime number using function recursion.

Q3.You are given a large integer represented as an integer array digits, where each digits[i] is the ith digit of the integer. The digits are ordered from most significant to least significant in left-to-right order. The large integer does not contain any leading 0's.

Increment the large integer by one and return *the resulting array of digits*.

Example :

Input: digits = [1,2,3] **Output:** [1,2,4]

Explanation: The array represents the integer 123.

Incrementing by one gives $123 + 1 = 124$.

Thus, the result should be [1,2,4].

Q4. Create a class called "Player" with members jersyno, name, run, and matches. Write a java program to store the input data for five player, display their all player information, and delete record in specific index then display record ascending orders by matches.

**Q5. Write a java program to create a class name as Employee using pojo class to accept id, name, email, sal and create method name as getFunction() to implement the logic.
Note : - Display Second Highest Salary Employee Record.**

Q6. Write a java program using constructor overloading with class name as ArrayCons. ArrayCons (int a [],int value): this function is used for accept array as parameter and pass value and search value in array.

ArrayCons(char a[]): this function will accept array as parameter and calculate sum of digit in string.

// Input : - giris123tech45hub67

// Output : - 123+45+67 = 235

**Q7. Write a java to Create class name as Food with using data member fid,fname, fprice,fcategory using do while and switch case.
create array of object of size 5 ,store the food details in it and perform following operations.**

case 1 : Add All Food details.

case 2 : Display All Food details.

case 3 : Display Bill Details :

Case 1 : Bill Without Gst.

Case 2 : Bill With 20% Gst.

case 4 : Display Food Details Start first letter is 'P' and ends with 'A'.

Q8. You are designing a banking system where different types of bank accounts share some common functionalities but have different ways of computing interest.

Task:

1. Create a base class BankAccount with the following:

- **accountNumber (String), balance (double) as instance variables.**
- **A constructor to initialize these fields.**
- **A method deposit(double amount) to add money to the account.**
- **A method withdraw(double amount) to deduct money (only if sufficient balance is available).**
- **A method displayDetails() to print account details.**

2. Create two subclasses:

- **SavingsAccount** (Interest is 4% per year, cannot withdraw more than 50% of balance).
- **CurrentAccount** (No interest, but allows overdraft up to ₹5000).

3. Override the withdraw() method in both subclasses to enforce these rules.

4. Write a main method to test the functionality by creating instances of both account types, depositing money, withdrawing it, and displaying account details.

Q9. You are designing a Mobile Wallet Payment System where different wallet services process payments differently.

Task:

1. Create an abstract class WalletPayment with:

- **walletID (String) and balance (double)** as instance variables.
- **A constructor** to initialize these fields.
- **An abstract method makePayment(double amount).**
- **A non-abstract method displayWalletDetails()** to print the wallet ID and balance.

2. Create two subclasses:

- **PaytmWallet:** Deducts 1.5% transaction fee from the amount before processing.
- **GooglePayWallet:** Deducts 1.2% transaction fee from the amount before processing.

3. Implement makePayment(double amount) in both subclasses to check if sufficient balance exists after the fee deduction and process the payment accordingly.

4. In the main method, create instances of PaytmWallet and GooglePayWallet, attempt payments, and display wallet details.

-----ALL THE BEST-----